

2022 - Fall

1.a) Why data communication is important? Draw and explain a generic data communication model.

sol → Data communication is the process of transferring data from one place to another or between two locations.

→ Data communication is important because data communication allows electronic and digital data to move between two networks, no matter where the two are located geographically, what the data contains, or what format they are in.

→ A common example of data communication is connecting your laptop to a ~~Wi~~ Wi-Fi Network.

A generic data communication model is shown below:

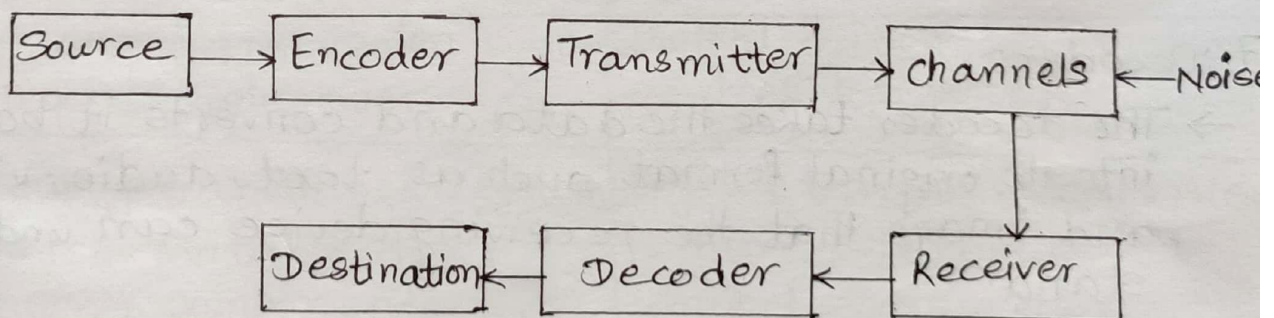


Fig. Generic data communication model

1. Source:

→ The source is that where the data originates, it can be a computer, phone or any device generating information.

2. Encoder:

→ The encoder takes the data, which can be in various format such as text, audio, video and image and converts it into a format suitable for transmission.

3. Transmitter:

→ Transmitter modulates the digital data into a signal that can be transmitted over a physical medium.

4. Channels:

→ The channel is the physical path the signal travels on, like cables, telephone lines, or even radio waves.

5. Noise:

→ The noise is an unwanted signal added to the original signal for the transmission of data for upto 100 Kms.

6. Receiver:

→ The receiver picks up the transmitted signal and converts it back into a digital format.

7. Decoder:

→ The decoder takes the data and converts it back into its original format such as text, audio, video and image that the receiving device can understand.

8. Destination:

→ The destination is the device that receives the processed data.

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1.a) How is digital transmission superior to analog transmission system?

sol Digital transmission is superior to analog transmission system because of the following differences:

S.N.	Analog Transmission System	Digital Transmission System
i>	Analog transmission system uses analog signals for transmission.	Digital transmission system uses digital signals (0's & 1's) for transmission.
ii>	Analog transmission system has high error probability due to parallax.	Digital transmission system has low error probability.
iii>	Analog transmission system requires low bandwidth.	Digital transmission system requires high bandwidth.
iv>	Analog transmission system has complicated and less flexible hardware.	Digital transmission system has less complicated and flexible hardware.
v>	Analog transmission system requires high power.	Digital transmission system requires low power.
vi>	Analog transmission system is less portable due to heavy components.	Digital transmission system is more portable due to compact equipments.
vii>	In analog transmission system, circuits are complicated to design.	In digital transmission system, circuits are easy to design.
viii>	In analog transmission system, the capacity of the channel is not effectively utilized.	In digital transmission system, the capacity of the channel is effectively utilized.

1.a) Write unique features of digital transmission system
eg The unique features of digital transmission system are as follows:

i) Rich media communication:

→ Digital transmission system allows for the transmission of various content types such as text, audio, video and images.

ii) Asynchronous and Synchronous communication:

→ Digital transmission system offers both synchronous and asynchronous communication options.

iii) Instantaneous and global reach:

→ Digital transmission system transcends geographical barriers.

iv) Searchability and archiving:

→ Digital transmission system creates a searchable archive of information.

v) Scalability and reach:

→ Digital transmission system can reach a vast audience with minimal effort.

vi) Accessibility:

→ Digital transmission system can be adopted to meet the needs of people with disabilities.

1.a) Explain the applications of data communication.

eg The applications of data communication are as follows:-

i) Computer Networks:

→ Data communication is the lifeblood of computer networks, enabling devices like computers, printers and servers to connect and share resources like files, applications and storage.

ii) Internet:

→ Internet is a global network of interconnected computer networks that facilitates data communication on a massive scale.

iii) World Wide Web (WWW):

→ WWW utilizes data communication to deliver web pages, images and other content to our devices.

iv) Email:

→ Email uses data communication to send and receive messages between individuals or groups.

v) Video Conferencing:

→ Video Conferencing uses data communication to transmit real-time audio and video between devices.

vi) Instant Messaging:

→ Instant messaging allows for real-time text-based communication between users.

vii) Online Gaming:

→ Online gaming relies on data communication to connect players and facilitate real-time interaction with a virtual environment.

1.a) What are the data communication terminologies?
Write about evolution of data communication system

The data communication terminologies are listed below:

i) Data channel:

→ The data channel is a computer path or medium used to transfer the data from sender to receiver.

ii) Baud:

→ Baud measures the rate at which the data is transferred from the sender to the receiver in the data channel.

iii) Bits Per Second (bps):

→ Bits Per Second is also a measurement unit of speed at which the data is transferred. Its symbol is "bit/s".

iv) Bandwidth:

→ Bandwidth is the difference between the highest and the lowest frequencies of a data channel.

v) DTR (Data Transfer Rates):

→ DTR determines the rate of speed at which the data is transferred from one point to another.

vi) Throughput:

→ Throughput calculates the network efficiency, by measuring data transfer between the source and destination over the network.

* Evolution of Data Communication System:

- In 1837, Samuel Morse exhibited a working Telegraph.
- In 1843, Alexander Bain patented a Printing Telegraph.
- In 1876, Alexander Graham Bell introduced the Telephone.
- In 1895, Guglielmo Marconi invented The Radio.
- In 1947, Bells Labs introduced The Transistor.
- In 1958, US-government ~~lana~~ launched Communication oriented satellite.
- In 1962, Fax Transmission was introduced.
- In 1969, Internet Protocol (IP) was developed.
- In 1991, World Wide Web (WWW) was emerged as the primary component of the Internet.

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1.6) Differentiate between different types of transmission modes. Briefly explain RS-232 C.

sol The differences between different types of transmission modes are shown below:

S.N.	Simplex Mode	Half-Duplex Mode	Full-Duplex Mode
i>	Simplex mode has unidirectional communication.	Half-Duplex mode has two directional communication one at a time.	Full-Duplex mode has two directional communication simultaneously.
ii>	In simplex mode, the sender can only send data.	In half duplex mode, the sender can send and receive data, but one at a time.	In full-duplex mode, the sender can send and receive data simultaneously.
iii>	In simplex mode, one channel usage for the transmission of data.	In half-duplex mode, one channel usage for transmission of data.	In full duplex mode, or two channels usage for transmission of data.
iv>	Simplex mode has less performance than half-duplex and full-duplex modes.	Half-duplex mode has less performance than full-duplex mode.	Half Full-duplex mode has better formance than simplex and half-duplex mode.
v>	Examples: Keyboard and Monitor.	Examples: walkie-Talkies.	Examples: Telephone.

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RS-232C:

- RS-232 is a standard protocol used for serial communication.
- RS-232 is a standard physical interface that defines electrical characteristics and mechanical specifications for connection a DTE ~~at~~ to a DCE.
- RS-232 uses a 9-pin connector and 25 signals lines of which 20 are used for data communication.
- Signal lines are divided into three categories:
 - i> Data lines; Data lines are used to transmit and receive signals.
 - ii> Control lines; Control lines are used to control the flow of data.
 - iii> Status lines; Status lines provide information about the status of data communication.
- The RS-232 standards are used at DTE and DCE.
- DTE (Data Terminal Equipment) produces data and transfers them to a DCE with essential control characters.
- DCE (Data Communication Equipment) converts signals to a format appropriate to transmission medium.
- DTE is connected through help of a DCE network.
- DCE network acts as a medium for two DTE networks.

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7. a) Write short notes on "Baud rate";

→ Baud rate is the rate at which the number of signal elements or changes to the signal occurs per second when it passes through a transmission medium.

$$\Rightarrow \text{i.e. } \underline{\text{Baud rate}} = \frac{\text{number of signal elements}}{\text{total time (in seconds)}}$$

- Higher the baud rate, faster the data is sent/received.
- Baud rate can determine the bandwidth requirements for transmission of the signal.
- Baud rate is also used for the calculation of the Bit rate of a communication channel.
- Baud rate is a turning parameter for the transmission of a signal.
- Baud rate specifies how fast data can be sent over a serial line or serial interface.

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1.6> Differentiate between serial and parallel communication.

sol

Serial Communication	Parallel Communication
i> In serial communication, a single link transfers data from one point-to-another.	In parallel communication, multiple links transfer data from one point-to-another.
ii> Serial communication is slow.	Parallel communication is fast.
iii> Serial Communication is used for long distance.	Parallel communication is used for short distance.
iv> In serial communication, only one bit is transferred at one clock pulse.	In parallel communication, 8 bits are transferred at one clock pulse.
v> Serial Communication is full duplex.	Parallel Communication is half duplex.
vi> Serial communication is cost efficient.	Parallel Communication is not cost-efficient.
vii> In serial communication, data flows in bi-direction.	In parallel communication, data flows in multiple lines.
viii> The circuit used in serial communication is simple.	The circuit used in parallel communication is complex.

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2.a) Explain Null-modem configuration in RS-232C. Explain the relationship between baud rate and bit rate.

Ans:

- Null modem configuration in RS-232C refers to the direct connection between two data terminal equipment (DTE) devices without the need for a modem.
- This setup is facilitated by using a null modem cable which crosses over the transmit (TX) and receive (RX) lines between the two devices.
- By using a null modem cable, the two DTE devices can establish bidirectional communication without the need for a modem.
- This setup is commonly used for direct connections between computers, terminals or other similar devices, facilitating data exchange and communication between them.
- Null modem configurations are particularly useful in scenarios where direct communication between devices is needed, such as file transfer, data logging or device control applications.

Explain

The relationship between baud rate and bit rate:

- Both bit rate and baud rate are generally used in data communication to measure the speed of data.
- Baud rate is defined as the number of signal units per second.
- Bit rate is defined as the transmission of a number of bits per second.
- The formula that relates both bit rate and baud rate is shown below:

$$\text{Bit rate} = \text{Baud rate} \times \text{the number of bits per baud}$$

2015-16
7.b) Write short notes on "Unguided Transmission Media".

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- An unguided transmission media is the transmission media that transmits the electromagnetic waves without using any physical medium.
 - Therefore, unguided transmission media is also known as wireless transmission media.
 - In unguided media, the signal energy communicates through the air.
 - An unguided media is generally preferred for radio broadcasting in all directions.
 - The unguided media formed the continuous network topologies.
 - Some unguided transmission media are listed below:
 - i> Microwave Transmission
 - ii> Radio Transmission
 - iii> Infrared Transmission

1. b) Explain the RS-232C standards with handshake signals.

sol:

- RS-232C is a standard protocol used for serial communication.
- RS-232C is a standard physical interface that defines electrical characteristics and mechanical specifications for connection a DTE and DCE.
- DTE stands for Data Terminal Equipment and DCE stands for Data Communication Equipment.
- RS-232C uses a 9-pin connector and 25 signal lines of which 20 are used for data communication.
- Handshaking is the process of transmitting signals from DTE to DCE in order to make connections before the actual data transfer.
- Following is the sequence of handshaking:
 - Initially, the computer activates RTS signal to modem when a data is transferred from computer to modem.
 - Modem in turn activates the DCD and then the CTS gets activated.
 - Computer then sends data on TXD. After the data transmission is completed, the computer deactivates the RTS which causes the modem to deactivate CTS.

1.a) Differentiate between synchronous and asynchronous communication.

sol

S.N.	Synchronous Communication	Asynchronous Communication
i)	In synchronous communication, data is transmitted in the form of blocks.	In asynchronous communication, data is transmitted in the form of bytes or characters.
ii)	Synchronous Communication is fast.	Asynchronous communication is slow.
iii)	Synchronous communication is costly.	Asynchronous communication is economical.
iv)	Synchronous communication needs precisely synchronized clocks.	Asynchronous communication need does not need precisely synchronized clocks.
v)	In synchronous Communication, the time interval of communication is constant.	In asynchronous communication, the time interval of communication is random.
vi)	In synchronous communication, there is no gap present between data.	In asynchronous communication, there is gap present between data.
vii)	In synchronous communication, errors are detected and corrected in real time.	In asynchronous communication, errors are detected and corrected, when the data is received.

1.6) Discuss different types of serial data transmission.
Also illustrate a situation when baud rate can be lower than bit rate.

Serial data transmission is a type of data transmission in which a single link transfers data from one point-to-another.

→ There are two types of serial data transmission:

i) Synchronous data transmission

ii) Asynchronous data transmission

* Next Part:

→ Baud rate is defined as the number of signal units per second. Also, baud rate is the number of times a signal in a communications channel changes state. For example; a 7000 baud rate means that the channel can change states up to 7000 times per second.

→ Bit rate is a measure of the number of data bits (i.e. 0's & 1's) transmitted in one second. For example 7000 bits per second means 7000 zeros or ones can be transmitted in one second.

→ In most of the cases, baud rate will be the same as the bit rate.

→ There are cases though where a channel can send 4 bits per baud, meaning that for every 4 bits, we have one change and in this case, the baud rate is $\frac{1}{4}$ th of the bit rate.

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2.a) Define LTI system, Explain with examples, how periodic and non-periodic signals are distinguished.

→ LTI stands for Linear Time Invariant.

→ LTI system is a type of system, if a system is both linear and time Invariant.

* Differences between periodic and non-periodic signals:

Periodic Signals	Non-Periodic signals
i) Periodic signal is a signal which repeats itself after a specific interval of time.	Non-periodic signal is a signal which does not repeat itself after a specific interval of time.
ii) Periodic signal repeats its pattern over a period.	Non-periodic signal does not repeat its pattern over a period.
iii) Periodic signal is represented by a mathematical eq ⁿ .	Non-periodic signal is not represented by any mathematical eq ⁿ .
iv) The value of periodic signal can be determined at any point of time.	The value of non-periodic signal cannot be determined at any point of time.
v) Periodic signals are deterministic signals.	Non-periodic signals are random signals.
vi) Periodic signals are: sine, cosine, square etc.	Non-periodic signals are: sound signals from radio etc.

2. b) Check whether $x(t) = A \cos \omega t$ is power or energy type signal.

solⁿ Here,

$$x(t) = A \cos \omega t$$

$$|x(t)|^2 = A^2 \cos^2 \omega t = A^2 \frac{(1 + \cos 2\omega t)}{2}$$

$$= \frac{A^2}{2} (1 + \cos 2\omega t)$$

The power signal is given by

$$\text{Power, } P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \frac{A^2}{2} (1 + \cos 2\omega t) dt$$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{2T} \int_{-T/2}^{T/2} (1 + \cos 2\omega t) dt$$

$$= \frac{A^2}{2} \lim_{T \rightarrow \infty} \frac{1}{T} \left[t + \frac{\sin 2\omega t}{2\omega} \right]_{-T/2}^{T/2}$$

$$= \frac{A^2}{2} \lim_{T \rightarrow \infty} \frac{1}{T} \left[T/2 + T/2 + \frac{1}{2\omega} (\sin \omega T + \sin \omega T) \right]$$

$$= \frac{A^2}{2} \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{2T}{2} \right]$$

$$= \frac{A^2}{2} \lim_{T \rightarrow \infty} (1)$$

$$\therefore \text{Power, } P = \frac{A^2}{2}$$

Also,

The energy signal is given by

$$\begin{aligned}\text{Energy, } E &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\&= \int_{-\infty}^{\infty} \frac{A^2}{2} (1 + \cos 2\omega t) dt \\&= \frac{A^2}{2} \int_{-\infty}^{\infty} \left[1 + \frac{\sin 2\omega t}{2\omega} \right] dt \\&= \frac{A^2}{2} \left[t + \frac{\sin 2\omega t}{2\omega} \right]_{-\infty}^{\infty} \\&= \frac{A^2}{2} (\infty + \infty)\end{aligned}$$

$$\therefore \text{Energy, } E = \infty$$

Since the given signal has finite power and infinite energy. So the given signal is power signal.

2.a) Define signal. Briefly, explain periodic and non-periodic signals, energy and power signals, deterministic and random signals, continuous and discrete time signals.

→ A signal is an electromagnetic or electrical current that carries data from one system or network to another.

i) Periodic and Non-periodic signals:

→ A periodic signal is a signal which repeats its pattern over a period.

→ A non-periodic signal is a signal which does not repeat its pattern over a period.

ii) Energy and Power Signals:

→ A energy signal is a signal which has finite energy and zero average power.

→ A power signal is a signal which has non-zero power and infinite energy.

iii) Deterministic and Random signals:

→ A signal is said to be deterministic signal if there is no uncertainty in determining its value at any time.

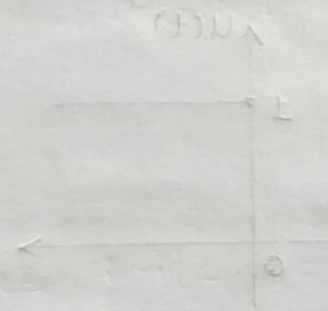
→ A signal is said to be random signal if there is some degree of uncertainty in determining its value at any time.

iv) Continuous and Discrete Time signals:

→ A continuous time signal is a signal that can be defined and represented at any time instant in the sequence.

→ A discrete time signal is a signal that can be defined and represented at certain time instants of the sequence.

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$



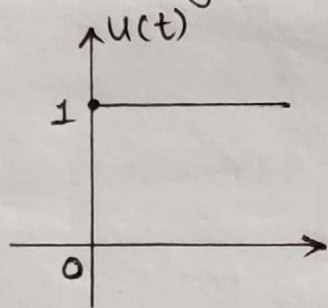
$$u[n] = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$



2. b) Define unit step signal with necessary mathematical expression and graphical illustration. Check the following system for Linearity, causality, static & dynamic and time invariance. $y(n) = Ax(n) + B$

sol

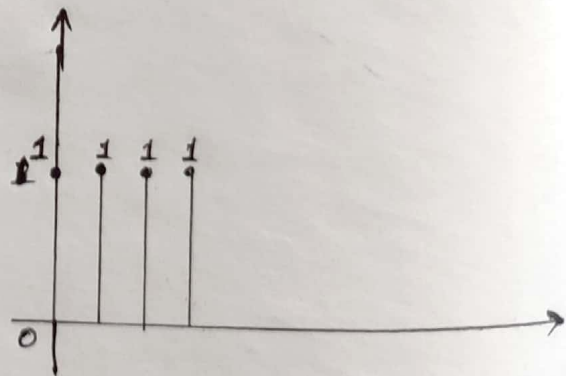
- A unit step signal is a signal whose amplitude is unity for the zero or positive value of time t and zero value for negative values of time t .
- A continuous time unit step signal is represented by $u(t)$ and is given by



$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

- A discrete time unit step signal is represented by $u(n)$ and is given by

$$u(n) = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$



2. a) Differentiate between energy and power signal with example. Justify whether $x(t) = e^{-a|t|}$ ($a > 0$) is energy or power signal.

soln
e.n.

Energy Signal	Power Signal
i) Total normalized energy is finite and non-zero.	The normalized average power is finite & non-zero.
ii) The energy is obtained by $E = \int_{-\infty}^{\infty} x(t) ^2 dt$	The average power is obtained by $T/2$ $P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) ^2 dt$
iii) Non-periodic signals are energy signals.	Periodic signals are power signals.
iv) Energy signals are limited time limited.	Power signals can exist over infinite time.
v) Power of energy signal is zero.	Energy of the power signal is infinite.

Here,

$$x(t) = e^{-a|t|} \quad (a > 0)$$

$$\therefore |x(t)|^2 = |e^{-a|t|}|^2 = e^{-2a|t|}$$

The energy signal is given by

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^{\infty} e^{-2a|t|} dt = \left[\frac{e^{-2at}}{-2a} \right]_0^{\infty}$$

$$= -\frac{1}{2a} (e^{-\infty} - e^{-0})$$

$$= -\frac{1}{2a} (0 - 1)$$

$$\therefore E = \frac{1}{2a}$$

Similarly,

The average power of a signal is given by

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} e^{-2at} dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{e^{-2at}}{-2a} \right]_0^{T/2}$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{e^{-aT} - e^0}{-2a} \right]$$

$$= -\frac{1}{2a} \lim_{T \rightarrow \infty} \frac{1}{T} \times e^{-aT}$$

$$\therefore P = -\frac{1}{2a} \times \frac{1}{\infty}$$

$$\therefore P = 0$$

Since the energy is finite and average power is zero, hence the given signal is energy signal.

2.a) Define Time Invariant system. Explain the following properties of the system; a) Causality, b) linearity, c) stability, d) memory.

sol

→ A system is said to be Time Invariant system if its input and output characteristics do not vary with time.

i) Causality:

→ A system is said to be causal if its output depends upon present and past inputs and does not depend upon future input.

ii) Linearity:

→ A system is said to be linear if it satisfies superposition and homogeneity principles.

iii) Stability:

→ A system is said to be stable if and only if every bounded input produces a bounded output.

iv) Memory:

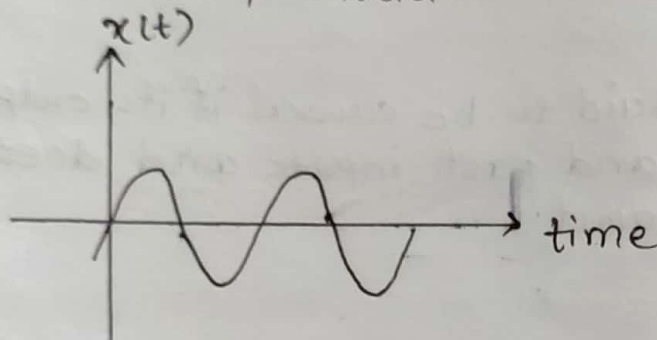
→ Memory is an electronic holding place for the instructions and data a computer needs to reach quickly.

2. b) Explain deterministic and random signal with examples. Justify whether unit step signal is energy signal or power signal.

sol

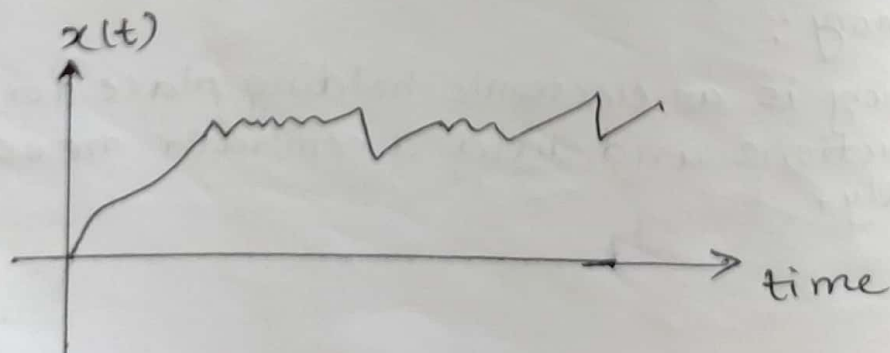
* Deterministic signal:

- A signal is said to be deterministic signal if there is no uncertainty in determining its value at any time.
- These signals can be represented by some mathematical formula.



* Non-Random Signal:

- A signal is said to be random signal if there is some degree of certainty in determining its value at any time.
- These signals cannot be described by a mathematical equation.
- They are modelled in probabilistic terms.



A unit step signal is a signal whose amplitude is unity for the zero or positive value of time t and zero value for negative values of time t .

i.e.

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

Now, The average power is given by

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} 1 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} [t]_{-T/2}^{T/2}$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} (T/2)$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2}$$

$$\therefore P = \frac{1}{2}$$

The energy is given by

$$E = \lim_{T \rightarrow \infty} \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \int_{-\infty}^{\infty} 1 dt$$

$$= \lim_{T \rightarrow \infty} [t]_{-\infty}^{\infty}$$

$$= \lim_{T \rightarrow \infty} (\infty)$$

$$\therefore E = \infty$$

Since the average power is finite and energy is infinite, so the unit step signal is power signal.

2.a) Find the energy and power of signal $x(t) = e^{-4t}u(t)$ and justify whether it is energy signal or power signal.

sg Here,

$$x(t) = e^{-4t}u(t)$$

$$\therefore |x(t)|^2 = e^{-8t}u(t)$$

The unit step signal $u(t)$ is given by

$$u(t) = \begin{cases} 1 & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$

The energy of a signal is given by

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^{\infty} e^{-8t} dt$$

$$= e^{-8t} \left[t \right]_0^{\infty}$$

$$= e^{-\infty} (\infty - 0)$$

$$\therefore E = \infty \quad \text{Ans}$$

The power of a signal is given by

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} e^{-8t} dt$$

$$= \lim_{T \rightarrow \infty} \frac{e^{-8t}}{T} \left[t \right]_0^{T/2}$$

$$= \lim_{T \rightarrow \infty} \frac{e^{-8 \cdot T/2}}{T} \times T/2$$

$$\therefore P = \lim_{T \rightarrow \infty} \frac{e^{-8}}{2} = \frac{e^{-8}}{2}$$

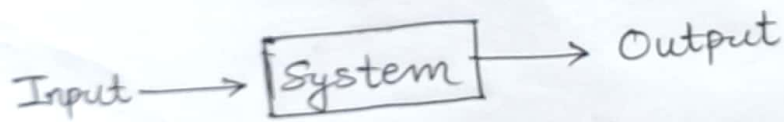
Since the energy of a signal is infinite and average power of a signal is finite, so the given signal is power signal.

(2018-Spring)

2. b) What is a system? How would you say that a system is BIBO stable? Comment about the stability for the following system.:

$$y(n) = 2x(n+1) + [x(n-1)]^2$$

→ A system is a set of elements which produces output in response to input.



→ The test of stability of the system is performed by applying specific input. On applying a bounded input to a system, if the output is bounded, then the system is said to be BIBO stable.

We have,

$$y(n) = 2x(n+1) + [x(n-1)]^2$$

2023-Fall (old)

3.a) What is Unit step Function and Impulse Function?
Obtain the fourier transform of Unit Impulse. Also comment on the result.

✱

*Unit step Function:

→ A unit step function is a type of elementary function which exists only for positive side and is zero for negative side.

→ It is denoted by $u(t)$ and is given by

$$u(t) = \begin{cases} 1 & ; t > 0 \\ 0 & ; t < 0 \end{cases} \text{ continuous usf}$$

$$u(n) = \begin{cases} 1 & ; n > 0 \\ 0 & ; n < 0 \end{cases} \text{ discrete usf}$$

*Unit Impulse Function:

→ A unit impulse function is a type of elementary function whose unit value exist only at time $t=0$.

→ It is denoted by $\delta(t)$ and is given by

$$\delta(t) = \begin{cases} 1 & ; t = 0 \\ 0 & ; \text{otherwise} \end{cases} \text{ continuous uif}$$

$$\delta(n) = \begin{cases} 1 & ; n = 0 \\ 0 & ; \text{otherwise} \end{cases} \text{ discrete uif}$$

The unit impulse function

$$\delta(t) = \begin{cases} 1 & ; t=0 \\ 0 & ; \text{otherwise} \end{cases}$$

The fourier transform of unit impulse function is given by

$$\begin{aligned} F[\delta(t)] &= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} 1 \cdot e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} e^{-j\omega t} dt \\ &= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\infty}^{\infty} \\ &= -\frac{1}{j\omega} (e^{\infty} - e^{-\infty}) \\ &= \frac{1}{j\omega} \end{aligned}$$

$$F[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$$= \left| 1 \cdot e^{-j\omega t} \right|_{t=0}$$

$$= e^{-j\omega \times 0}$$

$$= e^0$$

$$\therefore F[\delta(t)] = 1$$

Hence, the fourier transform of unit impulse function is

1. Ans

3.a) Explain the existence of Fourier transform. Find the Fourier transform of signum function.

~~3.a)~~ The Fourier transform does not exist for all non-periodic signal. Hence, for a function $x(t)$ to have Fourier transform, the following conditions should be satisfied:-

→ The function $x(t)$ is absolutely integrable over the time interval $(-\infty$ to $\infty)$ i.e.

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

→ The function $x(t)$ has a finite number of maxima and minima in every finite interval of time.

→ The function $x(t)$ has a finite number of discontinuities in every finite interval of time. Also each of these discontinuities must be finite.

The signum function is given by

$$\text{sign}(t) = \begin{cases} 1 & ; t > 0 \\ -1 & ; t < 0 \end{cases}$$

The Fourier transform of signum function is given by

$$\begin{aligned} F[\text{sign}(t)] &= \int_{-\infty}^{\infty} \text{sign}(t) \cdot e^{-j\omega t} dt \\ &= \int_{-\infty}^0 (-1) \cdot \text{sign}(t) dt + \int_0^{\infty} 1 \cdot \text{sign}(t) dt \end{aligned}$$

$$= \int_0^{\infty} 1 e^{-j\omega t} dt + \int_0^{\infty} 1 e^{-j\omega t} dt$$

$$= 2 \int_0^{\infty} e^{-j\omega t} dt$$

$$= 2 \times \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^{\infty}$$

$$= 2 \left(\frac{e^{-\infty} - e^0}{-j\omega} \right)$$

$$= \frac{-2}{-j\omega}$$

$$\therefore F[\text{sign}(t)] = \frac{2}{j\omega} = \frac{-2j}{\omega}$$

Hence, the fourier transform of signum function is $\frac{2}{j\omega} = \frac{-2j}{\omega}$. Ans

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2. b) State the properties of discrete time fourier series.
 Let $x(n)$ and $y(n)$ be two periodic signals with period N and fourier coefficients a_k and b_k respectively periodic with period N .

i) Linearity:

$$Z(n) = Ax(n) + By(n) \longleftrightarrow C_k = Aa_k + Bb_k$$

ii) Time shifting:

$$Z(n) = x(n-n_0) \longleftrightarrow C_k = a_k e^{-jk(2\pi/N)n_0}$$

iii) Time Reversal:

$$Z(n) = x(-n) \longleftrightarrow C_k = a_{-k} ; \begin{array}{l} a_k = a_{-k} \text{ for } x(n) \text{ odd} \\ a_k = -a_{-k} \text{ for } x(n) \text{ even} \end{array}$$

iv) Multiplication:

$$Z(n) = x(n)y(n) \longleftrightarrow C_k = \sum_{l \in \langle N \rangle} a_l b_{k-l}$$

v) Conjugation & Conjugate Symmetry:

- Real $x(n)$, $a_k = a_k^*$, (conjugate symmetric)
- Real and Even $x(n)$, $a_k = a_k^*$, (real & even a_k)
- Real and Odd $x(n)$, $a_k = -a_k^*$, (purely imaginary & odd a_k)
- $Z(n) = \text{Even part of } x(n) \longleftrightarrow C_k = \text{Real}\{a_k\}$
- $Z(n) = \text{Odd part of } x(n) \longleftrightarrow C_k = j \text{Imaginary}\{a_k\}$

vi) Periodic Convolution:

$$x(n) * y(n) = \sum_{r \in \langle N \rangle} x(r) y(n-r) \longleftrightarrow Na_k b_k$$

vii) Parseval's Relation:

$$\frac{1}{N} \sum_{n \in \mathbb{Z}N} |x[n]|^2 = \sum_{k \in \mathbb{Z}N} |a_k|^2$$

- Total average power = sum of average power in all harmonic components.
- Energy in time domain equals to energy in frequency domain.

2021-Fall

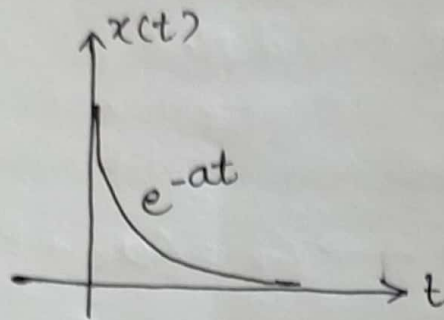
3. a) Why we need Fourier transform even we have Fourier Series?

~~3. a)~~ We need Fourier transform even we have Fourier Series because of the following reasons:-

- i) The Fourier transform extends the Fourier Series to handle non-periodic functions.
- ii) The Fourier transform analyzes signals over a continuous range of frequencies, essential for real-world applications.
- iii) The Fourier transform simplifies problems, especially with differential equations and convolutions.
- iv) The Fourier transform applies to functions over the entire real line, not just a specific interval.

3.6) Obtain the Fourier transform of a single sided exponential function of e^{-at} .

solⁿ The single sided exponential function, e^{-at} can be plotted as



The Fourier transform of e^{-at} is given by

$$F[x(t)] = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \frac{e^{-\infty} - e^0}{-(a+j\omega)}$$

$$= \frac{1}{a+j\omega}$$

$$\therefore F[x(t)] = \frac{1}{a+j\omega}$$

Hence, the Fourier transform of a single sided exponential function of e^{-at} is $\frac{1}{a+j\omega}$. Ans

chapter-5 Overview of Data Communication (3 hrs) Networking

2023-Fall (New)

3.a) Compare and contrast between OSI and TCP/IP with necessary figures.

OSI Layer	TCP/IP layer
i) OSI stands for Open System Interconnection.	TCP/IP stands for Transmission Control Protocol / Internet Protocol.
ii) OSI has 7 layers.	TCP/IP has 3 layers.
iii) OSI is less reliable.	TCP/IP is more reliable.
iv) OSI has strict boundaries.	TCP/IP does not have strict boundaries.
v) OSI follows a vertical approach.	TCP/IP follows a horizontal approach.
vi) OSI uses different session and presentation layers.	TCP/IP uses both session & presentation layers.
vii) OSI developed model then protocol.	TCP/IP developed protocol then model.
viii) In OSI layer, transport layer provides assurance delivery of packets.	In TCP/IP layer, transport layer does not provide assurance delivery of packets.
ix) In OSI layer, protocols are can be replaced easily.	In TCP/IP layer, protocols cannot be replaced easily.

3.6> Compare and contrast frame relay, X.25 and ATM.

Feature	Frame Relay	X.25	ATM
Data Rate	56 Kbps to 1544 Mbps	Up to 64 Kbps	25 Mbps to several Gps
Layer	Data Link (Layer 2)	Network (Layer 3)	Data Link (Layer 2)
Connection Type	PVC/SVC	PVC/SVC	VP/VC
Error Handling	Minimal	Extensive	Minimal
Overhead	Low	High	Moderate
Frame Size	Variable	Variable	Fixed
QoS Support	Limited	Limited	Extensive
Use Cases	Burst data traffic	Reliable, low-speed communication.	high-speed, low-latency transmission with strong QoS (Quality of Service).

3. b) List the different types of Network and Explain its topology.

So A computer network is a system in which multiple computers connected to each other to share information and resources.

The different types of Network are listed below:

i) Personal Area Network (PAN)

ii) Local Area Network (LAN)

iii) Metropolitan Area Network (MAN)

iv) Wide Area Network (WAN)

Network Topology:

1. Bus Topology:

→ In this topology, all the data that travels over the network is transmitted through a common transmission medium known as bus.

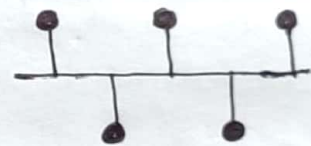


Fig. Bus Topology

2. Ring Topology:

→ In ring topology, every node in the network is connected to two other nodes and the first and the last nodes are connected to each other.



Fig. Ring Topology

3. Star Topology:

→ In star topology, each network node connected to a central node, which is known as a hub.

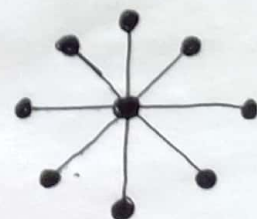


Fig. Star Topology

4. Mesh Topology:

→ In mesh topology, each network node is connected to every other node in the network.



Fig. Mesh Topology

5. Tree Topology:

→ In tree topology, the central root node is connected to one or more nodes of a lower hierarchy.

→ It is also known as hierarchical topology.

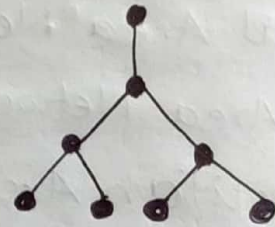


Fig. Tree Topology

4. a) ~~Di~~ State the role of CSMA/CD in data communication networking.

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→ CSMA/CD stands for Carrier Sense Multiple Access/ Collision Detection.

→ CSMA/CD is the protocol used in Ethernet to ensure that only one network node is transmitting on the network wire at any one time.

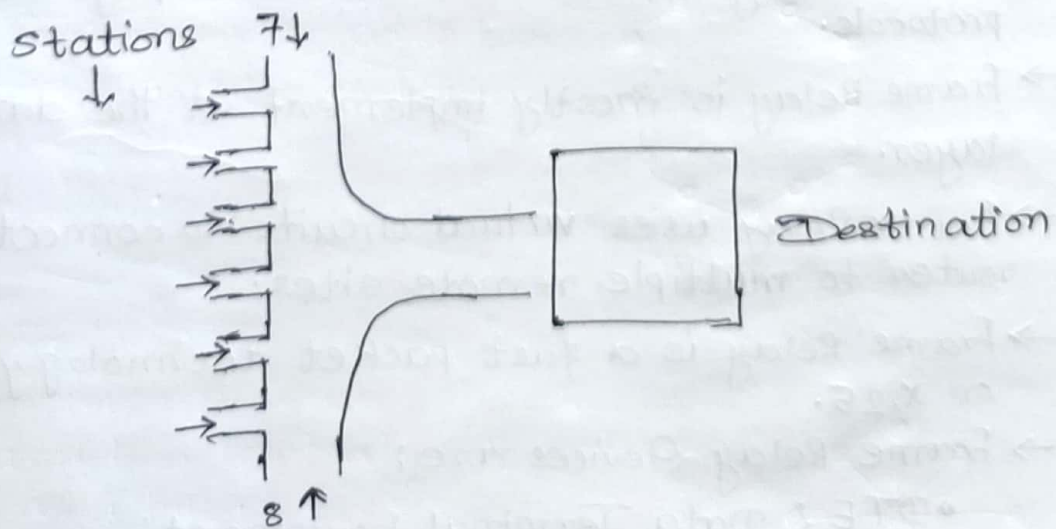
→ This is widely used in LANs.

→ For transmitting data through CSMA/CD protocol, CSMA/CD leads following rules of transmission:-

i) If the medium is idle, ~~otherwise~~ it transmits, otherwise go to step (ii).

ii) If the transmission medium or channel is busy, continue to listen ~~until~~ until the channel is idle then transmit immediately.

iii) If the collision is detected during transmission, it transmits a brief Jamming signal to assure all stations to know that there has been collision occurred and again transmit the same signal.



7. Write short notes on :

e) ATM and Frame Relay :

Ans

* ATM :

- ATM stands for Asynchronous Transfer Mode.
- ATM is a streamlined packet transfer interface.
- ATM makes use of fixed size packets called cells.
- In ATM, physical layer specifies transmission medium ~~between~~ and signal encoding scheme.
- In ATM, ATM layer defines transmission of data in fixed size cells and defines the use of logical connection.
- In ATM, ATM adaptation layer maps higher layer information ~~in to cells~~ into ATM cells to be transported over an ATM network.

* Frame Relay :

- Frame Relay is a packet switched communication service from LANs to backbone networks and WANs.
- Frame Relay operates at two layers: physical layer and data link layer.
- Frame Relay supports all standard physical layer protocols.
- Frame Relay is mostly implemented at the data link layer.
- Frame Relay uses virtual circuits to connect a single router to multiple remote sites.
- Frame Relay is a fast packet technology based on X.25.
- Frame Relay Devices are:
 - DTE : Data Terminal Equipment
 - DCE : Data Circuit Terminating Equipment

* X.25

- X.25 is a network protocol developed by ITU (International Telecommunication Union).
- X.25 is an ITU-T standard protocol suited for packet-switched data communication in WANs.
- X.25 specifies an interface between a host system and packet switching network.
- X.25 describes the procedures which are related for establishing, maintaining and terminating connections.
- The functionality of X.25 is specified on three layers:
 - i) Physical Layer:
 - Physical layer is concerned with electrical or signaling.
 - ii) Data Link Layer:
 - Data link layer is also known as frame layer.
 - iii) Packet Layer:
 - Packet layer governs the end-to-end communications among various DTE devices.

* Routing:

- Routing is the process of selecting paths in a network along with to send network traffic.
- Routing is the process of selecting the best path using some predetermined rules.
- Routing is performed by a device known as router.
- Routing is an autonomous process handled by the network devices to direct a data packet to its intended destination.
- Routing is a crucial mechanism that transmits data from one location to another across a network.
- Routing is of 3 types:
 - i> Static Routing:
 - static routing is also known as non-adaptive routing.
 - ii> Dynamic Routing:
 - Dynamic routing is also known as adaptive routing.
 - iii> Default Routing:
 - Default routing is commonly used when the network has single exit point.

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2023 - Fall (New)

3. b) Define guided and unguided transmission media. State the advantages of optical fiber over other guided media.

Sol

S.N.	Guided Transmission Media	Unguided Transmission Media
1.	Guided media is also known as wired communication media.	Unguided media is also known as wireless communication media.
2.	In guided media, the signal energy propagates through wires.	In unguided media, the signal energy propagates through the air.
3.	Guided media is used for point to point communication.	Unguided media is suited for radio broadcasting in all directions.
4.	Guided media forms discrete network topologies.	Unguided media forms continuous network topologies.
5.	Guided media is cost-effective.	Unguided media is expensive.
6.	In guided media, signals are in the form of voltage, current or photons.	In unguided media, signals are in the form of electromagnetic waves.
7.	Guided media is used for shorter distance.	Unguided media is used for longer distance.
8.	Some guided media are: twisted pair cables, co-axial cables & fiber optics cable.	Some unguided media are: microwave, radio wave, infrared wave etc.

→ An optical fiber is a type of guided transmission media that uses light to transmit data.

→ The advantages of optical fiber over other guided media are as follows:

i> Optical fibers can carry much larger amounts of data compared to copper cables.

ii> Optical fibers can transmit data over much longer distances without significant signal degradation.

iii> Optical fibers are not affected by electromagnetic interference (EMI).

iv> Optical fiber provides higher security for data transmission.

v> Optical fibers are thinner and lighter than copper cables, makes easy for handling and installing in tight spaces.

vi> Optical fiber does not corrode like metal cables, providing a longer lifespan.

vii> Optical fiber can operate effectively over a wide temperature range.

viii> Optical fiber provides clearer signals and lower latency.

ix> Optical fiber can easily support the increasing demands for higher data rates and network expansion.

4.a) Explain about different propagation techniques.
→ The different propagation techniques are discussed below:

i) Ground wave propagation:

- In ground wave propagation, radio waves travel through the lowest portion of the atmosphere.
- Ground wave propagation takes place below the frequency of 2 MHz.
- These low frequency signals radiate in all directions from transmitting antenna and follow the curvature of the earth.

ii) Sky wave propagation:

- In sky wave propagation, radio waves radiate up into the ionosphere and gets reflected back to the earth.
- Sky wave propagation takes place for frequencies between 2 - 30 MHz.
- Sky wave propagation allows communication at a greater distance.

iii) Line-Of-Sight (LOS) propagation:

- In LOS propagation, very high frequency signals are transmitted in straight line from antenna to antenna.
- Antennas must be directional and facing each other.
- LOS propagation takes place for frequencies above 30 MHz.
- Distance of signal propagation is limited due to the curvature of the earth in LOS propagation.

4.a) Compare the characteristics of twisted-pair cable and co-axial cable. Explain the working principle of satellite communication.

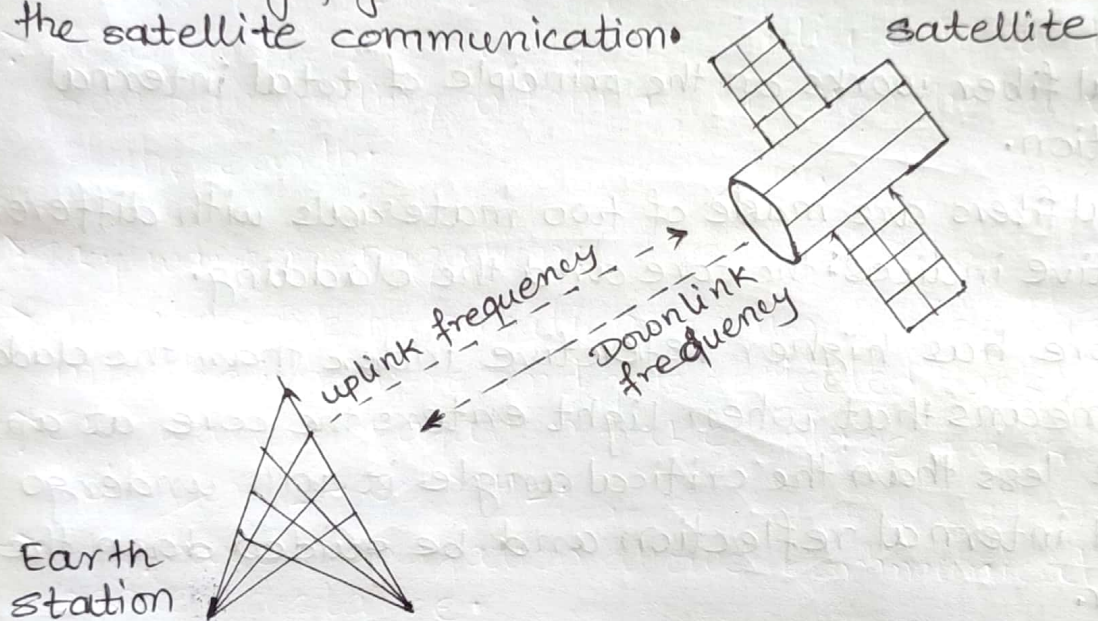
S.N: Twisted-Pair Cable	Co-axial Cable
i) In twisted-pair cable, the transmission of signals happens via metallic conducting wire.	In co-axial cable, the transmission of signals happens via the inner conductor of the wire.
ii) In twisted-pair cable, speed is up to 10 Gbps.	In co-axial cable, speed is up to 10 Mbps.
iii) Twisted-pair cables are easy to install and implement.	Co-axial cables are difficult to install and implement.
iv) Twisted-pair cable supports a bandwidth of low range.	Co-axial cable supports a bandwidth of moderate range.
v) Electromagnetic Interference (EMI) impacts twisted-pair cable.	Electromagnetic Interference (EMI) does not impact co-axial cable.
vi) Twisted-pair cable is less expensive.	Co-axial cable is more expensive.
vii) Types of Twisted-Pair cable: - Unshielded Twisted Pair (UTP) - Shielded Twisted Pair (STP)	Types of Co-axial cable: - RG-6 - RG-11 - RG-59

Satellite Communication:

- A satellite is a body that moves around another body in a mathematically predictable path called an orbit.
- A satellite communication is a microwave repeater station in space that is helpful in telecommunication, radio and television.

*Working Principle:

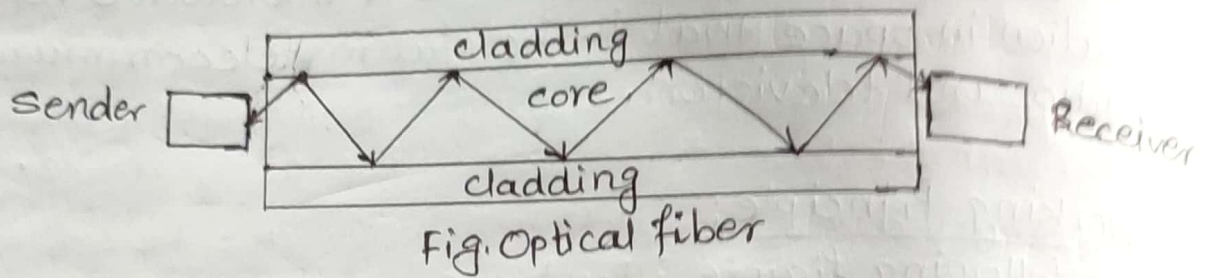
The following figure illustrate the satellite communication.



- A repeater is a circuit which increases the strength of the signal it receives and retransmits it.
- But here this repeater works as a transponder, which changes the frequency band of the transmitted signal, from the received one.
- The frequency with which the signal is sent into the space is called Uplink frequency.
- The frequency with which the signal is sent by the transponder is called Downlink frequency.

4.b) Explain the working principle of optical fiber.

* Working principle of optical fiber:



- Optical fiber works on the principle of total internal reflection.
- Optical fibers are made of two materials with different refractive indices: the core and the cladding.
- The core has higher refractive index than the cladding.
- This means that when light enters the core at an angle less than the critical angle it will undergo total internal reflection and be guided down the fiber.
- The light ray enters the fibre at an angle greater than the critical angle and is reflected back into the fibre at each interface between the core and the cladding.
- This process continues until the light ray reaches the other end of the fibre.

4.6> Compare guided transmission media in terms of bandwidth, channel capacity, SNR, security and cost.

sg Comparison of guided media:-

Features	Twisted-Pair	Co-axial	Optical fiber
Bandwidth	up to 2000 MHz	up to several GHz.	Hundreds of MHz to tens of THz.
Channel capacity	Up to 40 Gbps	Up to 10 Gbps	10 Gbps to 100 Tbps.
SNR (Signal-To-Noise Ratio)	Moderate	High	Excellent
Security	Moderate	High	Very High
cost	Low to moderate	Moderate	High (initial); decreasing
Performance	Suitable for home and office networks.	Suitable for high frequency signals	Suitable for long distance and high capacity.

7. Write short notes on:

a) Cellular Telephony:

sol:

- Cellular Telephony is designed to provide communication between two moving units called Mobile Stations.
- Cellular Telephony provides a wireless connection to the terrestrial telephone network for any user location within the radio range of the system.
- The architecture of cellular network is shown below where A, B, C, D, E, F and G are cells of hexagonal shape.

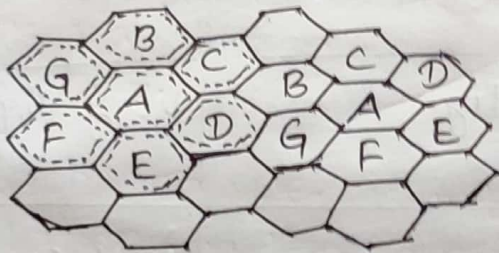


Fig. cellular Network with 7 cells

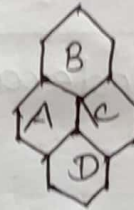


Fig. cellular Network with 4 cells

- Generally, cellular network is 7 cell or 4 cells structure and this structure is known as cluster.
- Each cell can cover large population, depending on population, the cell size can be designed for good signal coverage.
- In city area, cell size is small but in village or rural area, its size is large.
- All cells have different frequency in one 7 cell structure, this is for no occurrence of interference, crosstalk.

3.b) Compare twisted pair cable, co-axial cable and optical fiber based on frequency range, attenuation delay and repeater spacing.

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Property	Twisted-Pair	Co-axial	Fiber Optics
Frequency Range	Upto 250-600 MHz	Upto 10 GHz	From 186-194 THz
Attenuation	0.2-0.4 dB/meter	0.1-0.2 dB/meter	0.2 dB/km
Delay	200,000 kms^{-1}	230,000 kms^{-1}	200,000 kms^{-1}
Repeater Spacing	100 meters	1-2 Kilometers	40-100 Kilometers

7. Write short notes on:

a) VSAT:

~~809~~

- VSAT stands for Very Small Aperture Terminal.
- VSAT is a two-way ground station that transmits and receives data from satellites.
- A VSAT consists of two parts:
 - • A transceiver placed outdoors in direct line of sight to the satellite.
 - • A ^{device} placed indoors to interface transceiver with end user's communication device, such as a pc.
- The transceiver receives or sends a signal to a satellite transponder in the sky.
- The satellite sends and receives signals from a ground station computer that acts as a hub for the system.
- Each end user is interconnected with the hub station via the satellite, forming a star topology.
- The hub controls the entire operation of the network.

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